LUCAS FENG

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EDUCATION

Johns Hopkins University, Whiting School of Engineering — Expected May 2029
Bachelor of Science, Mechanical Engineering — Cumulative GPA: 3.88/4.0
Relevant Coursework: Statics and Mechanics of Materials, Mechanical Engineering Thermodynamics, Manufacturing Engineering

SKILLS

- **Design and Simulation:** SolidWorks (CAD & FEA), Onshape, KiCad
- **Languages:** Python, C++
- **Manufacturing:** CNC lathe/mill, FDM/resin 3D printing, laser cutting

EXPERIENCE

Mechanical Engineering Researcher in Dexterous Robotic Hands — Sep 2026 – Present
Johns Hopkins SciPhy Laboratory

- Developing the ODD hand, a human-sized, open-source robotic hand with quasi-direct-drive actuation, backdrivability, and force-torque sensing, designed to bring commercial-level dexterity to robot learning research at lower cost.
- Sourced multiple actuator candidates and characterized output torque, backdrivability, thermal performance, repeatability, backlash, and reflected inertia through bench testing to guide actuator selection and mechanical integration.

Mechanical Engineering Researcher in Force and Tactile Sensing — May 2026 – Present
Johns Hopkins Neuroengineering and Biomedical Instrumentation Laboratory

- Led end-to-end development of a low-cost 3-axis force sensor for measuring contact forces during robotic manipulation and calibrating tactile sensors, along with the mechanical design of a large-area tactile sensing system; presented the force sensor at IEEE IROS 2026.
- Developing a compact 6-axis force-torque sensor using PCB-integrated copper strain gauges, leading concept development, FEA, material selection, and PCB layout to enable lower-cost, fingertip-scale force sensing for robotic hands.

Drivetrain Engineer — Sep 2025 – Present
Johns Hopkins Baja SAE

- Disassembled and rebuilt the driveshaft and front differential of a Baja SAE competition vehicle as a member of a 5-person drivetrain subteam.
- Designed and tested a modular photogate base with independent leveling adjustments to maintain a level sensor position on uneven terrain.

Associate — Feb 2026 – Present
A-Level Capital

- Sourced ~40 founders and led introductory calls to assess team, product, and traction for a Johns Hopkins-affiliated, student-run venture fund managing $5 million in assets.
- Conducted due diligence on market, competition, and financials, building Excel models to project ARR growth and market size; synthesized findings into investment memos.

PROJECTS

3-Axis Force Sensor, *Johns Hopkins Neuroengineering and Biomedical Instrumentation Laboratory*

- Designed and fabricated a 3-axis force sensor at ~2% of the cost of commercial sensors, using FEA to maximize load transfer through four strain-gauge load cells and selecting electronics to provide high-rate force feedback for real-time robotic control.
- Iterated load-cell placement across 3 prototypes to improve load transfer and limit creep and hysteresis, then developed deadweight calibration and characterization procedures to recover three-dimensional forces from four load-cell outputs.
- Validated the sensor against a commercial reference across 340,000+ in-contact samples, achieving RMS error ≤ 0.12 N and beating published low-cost alternatives.

Robotic Tactile Skin, *Johns Hopkins Neuroengineering and Biomedical Instrumentation Laboratory*

- Designed silicone tactile bump unit cells that enable normal and shear sensing for large-area robotic tactile skin, owning SolidWorks geometry, mold design, silicone casting, and integration with the sensing hardware.
- Optimized bump height and profile geometry across 30 designs using nonlinear contact FEA, identifying a configuration that retained 98% of baseline shear gain at 68% of baseline deflection in simulation.
- Performed 50 grip-and-slip trials using sensor-equipped robotic grippers, analyzing synchronized tactile and vision data to support slip-prediction development.